

LESSON 11.9

REVISITING RELATIONSHIPS IN CIRCLES



LESSON INTRODUCTION

MA.7.GR.1.3 Explore the proportional relationship between circumferences and diameters of circles. Apply a formula for the circumference of a circle to solve mathematical and real-world problems.

MA.7.GR.1.4 Explore and apply a formula to find the area of a circle to solve mathematical and real-world problems.

LEARNING TARGETS

- I can find the circumference of a circle in terms of π .
- I can find the area of a circle in terms of π .
- I can relate proportional relationships to circles.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.GR.3.4 Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane involving perimeter or area of polygons.

MA.912.GR.4.4 Solve mathematical and real-world problems involving the area of two-dimensional figures.

ESSENTIAL QUESTIONS

- How can you relate proportional relationships to circles?

LESSON NARRATIVE

In this lesson, students start with a refresher on the parts of a circle and how they relate from Unit 4. Students will then explore which relationships in a circle are proportional. The lesson provides an opportunity to spiral content from an earlier unit while making new connections to the content learned in this unit. In the 11.9.4 Collaboration, students apply these skills to represent the proportional relationship between circumference and diameter in a variety of ways, including recognizing the constant of proportionality is π .

COMPONENT SUMMARY

11.9.1 WARM-UP (~10 MIN)

Refresher on circles

11.9.2 EXPLORATION (10-15 MIN)

Examine relationships in circles to determine which are proportional

11.9.3 BRING IT TOGETHER (5-10 MIN)

Summarize which relationships in circles are proportional

11.9.4 COLLABORATION (5-10 MIN)

Represent the proportional relationship between circumference and diameter in multiple ways

11.9.5 WRAP-UP (~5 MIN)

Reflection on how students felt they engaged with the lesson

RESOURCES

GLOSSARY

Key Terms:

- Area of a circle
- Circumference of a Circle

Additional Terms:

- Radius
- Diameter
- Constant of proportionality

MATERIALS

- N/A

ADDITIONAL PREPARATION

- 11.9.1 Warm-Up contains an optional video that is approximately one minute long.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may mix up which is the radius and which is the diameter when looking at a circle.
- Students may confuse which order to divide when finding constant of proportionality.

THINGS TO CONSIDER

- Consider drawing and labeling the different parts of the circle for students, if needed, for review.
- Consider writing the equations for area and circumference on the board for students to reference.
- If any anchor charts were created during Unit 4, consider displaying them again and adding to them with the new knowledge acquired.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

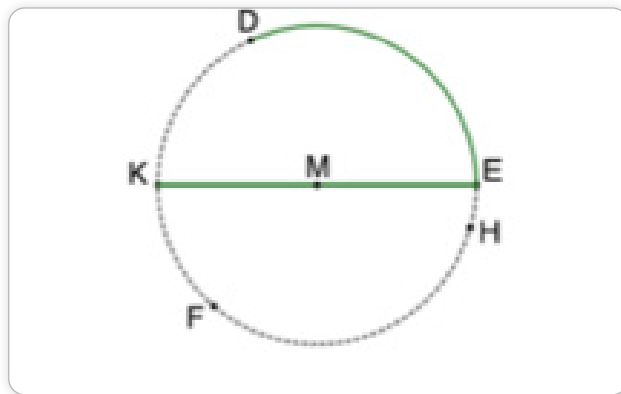
11.9.2 Collaboration positions students to engage in a Think-Pair-Share. Students have some time to look at the graphs independently before discussing with their partner and adjusting their work as necessary.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD**MA.K12.MTR.2.1 Multiple Representations**

Students' abilities to connect their knowledge of circles and apply it to what they have learned about proportional relationships demonstrates their abilities to express connections between multiple concepts. Specifically, in 11.9.2 Exploration, where students connect these different concepts while also translating between multiple representations of the given proportional or nonproportional relationships.

DIFFERENTIATE INSTRUCTION

This lesson provides a variety of visual entry points for students to engage with the content. The 11.9.1 Warm-Up positions students to activate prior knowledge of circles through color-coded diagrams, encouraging students to make connections between the different parts of the circle and how they are presented visually.

**11.9.1 WARM-UP: BELL WORK****TEACHER MOVES**

You can choose to play the optional video here to activate the 11.9.1 Warm-Up. This Warm-Up is meant to serve as Bell Work and a refresher from Unit 4. Students may need to spend a little extra time activating prior knowledge on circles.

**11.9.1 Warm-Up: Bell Work**

A series of screenshots of Circle M are shown. In each screenshot, the line segment and arc that are the same color have the same length.

Description	Screenshot
The diameter, KE , is wrapped around the circle starting at point E and ending at point D .	
The diameter, KE , is wrapped around the circle starting at point D and ending at point F .	
The diameter, KE , is wrapped around the circle starting at point F and ending at point H .	
A small part of Circle M 's circumference is shown between points H and E . Its equivalent length is shown on the diameter.	
All 3 diameters are shown wrapped around Circle M 's circumference, as well as the small amount of the circumference that is not a complete diameter.	

1. Complete the statements to describe the relationship shown in the series of screenshots of Circle M .

The ☐ perimeter
☒ circumference of any circle is equal to the length of the ☐ radius
☒ diameter times π . The value of π is approximately 3.14. The screenshots show the length of the ☐ perimeter
☒ circumference is the same as the length of slightly more than ☐ 2
☒ 3
☐ 4 ☐ radii.
☒ diameters.

2. The part of the circumference that represents the small amount of the circumference that is not a complete diameter has an approximate length of 0.14.

11.9.1 WARM-UP: BELL WORK (CONTINUED).

DIFFERENTIATE INSTRUCTION

If needed, consider drawing and labeling the different parts of the circle for students for review.

11.9.2 EXPLORATION: REVISITING RELATIONSHIPS IN CIRCLES

TEACHER MOVES

The 11.9.2 Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the 11.9.2 Exploration as they engage with each new problem.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group for 11.9.3 Bring It Together. Additional scaffolding questions are provided if needed, but this should be a primarily student-directed Exploration.



11.9.2 Exploration: Revisiting Relationships in Circles

The formulas for finding the circumference and area of a circle are shown.



Area of a Circle, A

$$A = \pi r^2, \text{ where } r \text{ is the radius}$$

Circumference of a Circle, C

$$C = \pi d \text{ or } C = 2\pi r,$$

where d is the diameter and r is the radius

1. Work with your partner to complete each table of values in terms of π .

Table A					
Diameter (inches)	1	2	3	4	5
Circumference (inches)	π	2π	3π	4π	5π

Table B					
Radius (inches)	1	2	3	4	5
Circumference (inches)	2π	4π	6π	8π	10π

Table C					
Radius (inches)	1	2	3	4	5
Area (sq. inches)	π	4π	9π	16π	25π

ENRICHMENT

Students may struggle to match each graph to the correct table.

- What is the scale of each axis? How can you use this to determine the matching table?
- Which values are plotted on the x-axis? Y-axis? How do you know?

TEACHER MOVES

Think-Pair-Share

Questions 3 and 4 position students to engage in a Think-Pair-Share. While students are discussing question 4, listen for and highlight responses that include mention of the graph being a straight line that goes through the origin. This is important in being able to reason about proportional relationships using graphs.

2. Three graphs are shown.
On your own, mark the table that matches each graph and label the x- and y-axes appropriately.

Graph	Table
	☐ Table A ☒ Table B ☐ Table C
	☐ Table A ☐ Table B ☒ Table C
	☒ Table A ☐ Table B ☐ Table C

3. Check your work with your partner and if necessary, resolve any differences.
Answers will vary.
4. Discuss with your partner which relationships appear to be proportional, including what evidence there is to support your thinking. Then, summarize your discussion.

Answers will vary. The graphs of table B and of table A are both proportional because the lines both go through the origin and form a straight line. The graph of the relationship between radius and area (table C) is not proportional because it is not a straight line.



11.9.3 Bring It Together: Revisiting Relationships in Circles

Complete the statements.

1. There ☐ is ☐ is not a proportional relationship between the diameter and circumference of a circle.

- ☐ Therefore, there is no constant of proportionality.
- ☐ The constant of proportionality is π .

2. There ☐ is ☐ is not a proportional relationship between the radius and circumference of a circle.

- ☐ Therefore, there is no constant of proportionality.
- ☐ The constant of proportionality is 2π .

3. There ☐ is ☐ is not a proportional relationship between the radius and area of a circle.

- ☐ Therefore, there is no constant of proportionality.
- ☐ The constant of proportionality is _____.

11.9.3 BRING IT TOGETHER: REVISITING RELATIONSHIPS IN CIRCLES

TEACHER MOVES

As you guide students through this section, give them time to consider each statement and discuss with a partner or small group. Use this section to encourage synthesis, not memorization, through strategic questioning:

- How do you know there is a proportional relationship between diameter/radius and circumference?
- Why isn't the relationship between the radius and area of a circle proportional? How can you show this?

11.9.4 COLLABORATION: DIAMETER AND CIRCUMFERENCE

TEACHER MOVES

As you guide students through this activity, give them time to create the table themselves and discuss with a partner or small group. Use this section to encourage synthesis, not memorization, through strategic questioning:

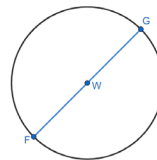
- What relationship can be determined for diameter/radius and circumference?
- Can a table be made swiftly if the column heads were “Radius” and “Circumference”?

TEACHER MOVES

Consider highlighting the relationship between the equation $c = \pi d$ and $y = \pi x$.

11.9.4 Collaboration: Diameter and Circumference

Circle W is shown with diameter FG .



1. Miranda uses a drawing program to change the size of Circle W . Find the circumference, in terms of π , for each diameter she uses.

Diameter	Circumference
12	12π
19	19π
45	45π
146	146π

2. Suppose x represents the diameter and y represents the circumference. Write the constant of proportionality, $\frac{y}{x}$.
 $\frac{146\pi}{146} = \pi$
3. Complete the statement.
 As the value of the diameter increases by 1, the value of circumference increases by π .
4. Write an equation that represents the relationship between x , the diameter, and y , the circumference.
 $y = \pi x$



11.9.5 Wrap-Up: Reflection

Take a **SELFIE** by checking the box that best matches what you did today.
Answers will vary.

Showed my work	☐	☐
Explained my answers	☐	☐
Lots of math vocabulary used	☐	☐
Found connections to other content	☐	☐
I persevered through the problem	☐	☐
Eliminated careless errors	☐	☐

11.9.5 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

Consider having students go back and put a star next to the problems that they think are proof of how they did on this lesson.